

CHE221 — Chemical Engineering Thermodynamics

Department of Chemical Engineering | Roadmap covering conservation principles, postulatory thermodynamics, potential transformations, solution thermodynamics, phase equilibria & reaction equilibrium

1 CORE FUNDAMENTALS & FIRST LAW ANALYSIS

1.1 System Classifications & Boundary Interactions

- **Open System** (control volume) – mass, momentum & energy cross the boundary; analyzed with continuity & energy transport balances.
- **Closed System** (control mass) – only heat & work cross the boundary; total mass invariant.
- **Isolated System** – no mass or energy exchange: $\Delta U = 0$, $\Delta S_{\text{universe}} \geq 0$.

1.2 State Functions vs. Path Functions

State functions (U, H, S, G, A, P, T, V) are exact differentials — path independent:

$$\int_1^2 dM = M_2 - M_1$$

Path functions (δQ , δW) are inexact differentials — depend on the path taken between states.

1.3 Energy Conservation Equations

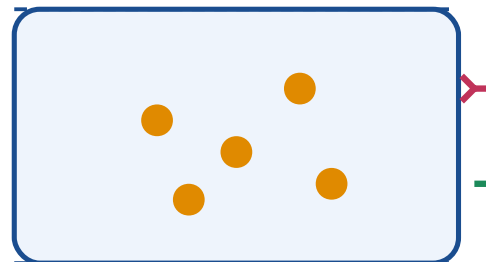
Closed system, differential form:

$$dU = \delta Q + \delta W_{\text{rev}} = \delta Q - P dV \quad (1)$$

Open system, steady-state flow balance:

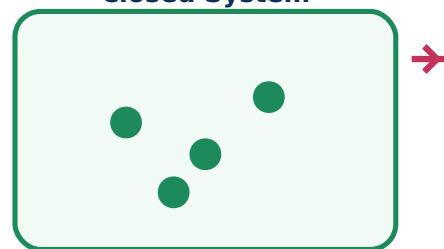
$$\Delta [(H + u^2/2 + gz) \dot{m}] = \dot{S} + \dot{t}_s \quad (2)$$

Open System



Control volume: mass, heat & work cross the boundary

Closed System



Mass fixed · only Q, W cross

Module 1 resources:

LearnChemE – First Law: Open & Closed Systems
MIT OCW – Energy Conservation & State Functions

Quick recap:

- Open → mass + energy cross boundary
- Closed → only Q, W cross boundary
- Isolated → nothing crosses boundary
- State fn: path-independent · Path fn: path-dependent

2 AXIOMATIC POSTULATES OF POSTULATORY THERMODYNAMICS

Postulate I – State Postulate: Simple, isotropic equilibrium states are completely and uniquely specified by U , V , and mole numbers $N_1, N_2, \dots, N_r$.

Postulate II – Entropy Maximization: A scalar function S exists for every equilibrium state. Absent internal constraints, extensive parameters take the values that maximize S over the constrained equilibrium manifold.

Postulate III – Additivity, Continuity & Monotonicity: $S = \sum S^{(k)}$; S is continuous, differentiable and strictly monotonic in U :

$$(\partial S / \partial U)_{V,N} = 1/T > 0 \quad (3)$$

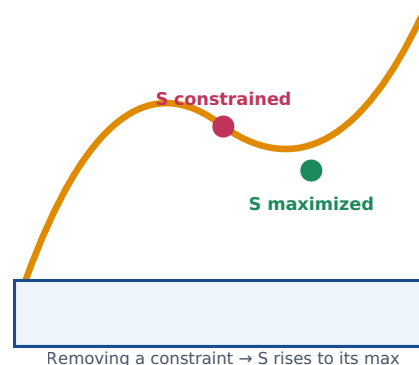
Postulate IV – Nernst / Third Law: Entropy of a pure crystalline system vanishes as $T \rightarrow 0$:

$$\lim_{T \rightarrow 0} S = 0, \text{ where } (\partial U / \partial S)_{V,N} = T \rightarrow 0 \quad (4)$$

Fundamental relation: Postulates I–III combine into the single equation $S = S(U, V, N_1, \dots, N_r)$ — every equilibrium thermodynamic property of a simple system can, in principle, be derived from it.

Consequence: Since $T = (\partial U / \partial S)_{V,N}$, temperature is *defined* from these postulates rather than assumed — this is what makes the framework “axiomatic” rather than empirical.

Entropy Maximization



Postulate	Core statement
I – State	(U, V, N_r) fix the equilibrium state
II – Max S	S is maximized at equilibrium
III – Additivity	S additive; $(\partial S / \partial U)_{V,N} = 1/T > 0$
IV – Nernst	$S \rightarrow 0$ as $T \rightarrow 0$ (pure crystal)

Module 2 resource:

NPTEL Thermodynamics – Axiomatic Postulates Framework

Why it matters: These four postulates replace the classical zeroth–third laws with a single self-consistent axiomatic base — every later result (Maxwell relations, equilibrium criteria, phase & reaction equilibrium) is derived from them.

3 THERMODYNAMIC POTENTIALS, MAXWELL RELATIONS & EOS

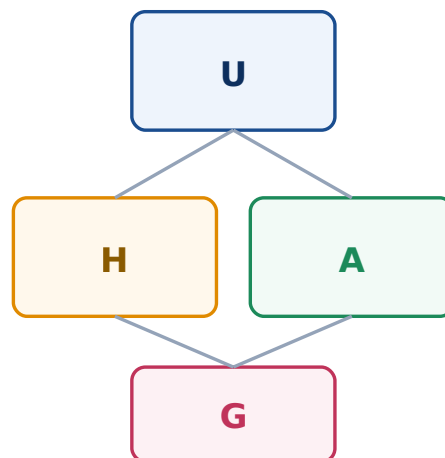
3.1 Fundamental Differential Property Relations

$$dU = TdS - PdV + \sum \mu_i dN_i$$

$$dH = TdS + VdP + \sum \mu_i dN_i$$

$$dA = -SdT - PdV + \sum \mu_i dN_i$$

$$dG = -SdT + VdP + \sum \mu_i dN_i \quad (5-8)$$



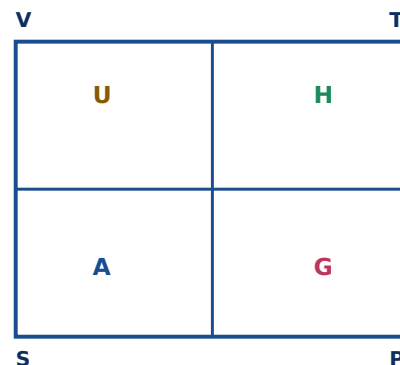
Legendre transforms linking U, H, A, G

3.2 Canonical Maxwell Relations

Derived from Euler reciprocity ($\partial^2 M / \partial x \partial y = \partial^2 M / \partial y \partial x$) applied to exact differentials:

$$(\partial T / \partial V)_S = -(\partial P / \partial S)_V \quad (\partial T / \partial P)_S = (\partial V / \partial S)_P$$

$$(\partial P / \partial T)_V = (\partial S / \partial V)_T \quad (\partial V / \partial T)_P = -(\partial S / \partial P)_T \quad (9-12)$$



Mnemonic square: corners = S,T,P,V · sides host U,H,A,G

Use: Maxwell relations replace hard-to-measure quantities like $(\partial S / \partial P)_T$ with PVT data — the backbone of residual-property and EOS calculations.

3.3 Residual Properties (M^R)

Quantify real-gas deviation from ideal gas at the same T, P:

$$M^R(T,P) = M(T,P) - M^{IG}(T,P)$$

Residual Gibbs energy:

$$G^R/RT = \int_0^P (Z-1) dP/P \quad Z = PV/RT \quad (13)$$

Module 3 resource: LearnChemE – Maxwell Relations & Residual Properties

4 SOLUTION THERMODYNAMICS & IDEAL GAS MIXTURES

4.1 Mathematical Framework of Partial Molar Properties

Definition:

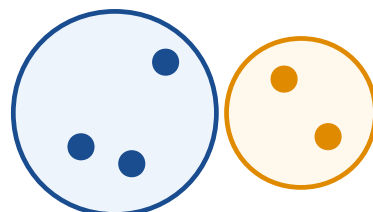
$$\bar{M}_i = \left(\partial M / \partial n_i \right)_{T,P,n_{j \neq i}} \quad (14)$$

Summation rule:

$$M = \sum_i x_i \bar{M}_i \quad (15)$$

Gibbs–Duhem (constant T, P):

$$\sum_i x_i d\bar{M}_i = 0 \Rightarrow \sum_i x_i d\mu_i = 0 \quad (16)$$



Binary mixture: species i, j

4.2 Property Changes of Mixing – Ideal Gas Mixtures

Property M	Partial molar M_i^{IGM}	$\Delta_{mix} M_i^{IGM}$
Internal Energy U	$U_i^{IG}(T)$	0
Enthalpy H	$H_i^{IG}(T,P)$	0
Volume V	$V_i^{IG}(T,P)$	0
Entropy S	$S_i^{IG}(T,P) - R \ln x_i$	$-R \sum x_i \ln x_i$
Gibbs Energy G	$G_i^{IG}(T,P) + RT \ln x_i$	$RT \sum x_i \ln x_i$
Helmholtz Energy A	$A_i^{IG}(T,P) + RT \ln x_i$	$RT \sum x_i \ln x_i$

Mixing of ideal gases is isothermal & isobaric – only S, G, A change; U, H, V are unaffected.

Module 4 resource: NPTEL Thermodynamics – Solution Thermodynamics & Partial Molar Properties

5 FUGACITY & INTERMOLECULAR FORCE MECHANICS

5.1 Phase Equilibrium & Fugacity

Equal chemical potentials across phases $\Rightarrow$ equal fugacities:

$$f_i^\alpha = f_i^\beta = \dots = f_i^\pi \quad (17)$$

5.2 Escaping Tendencies & Intermolecular Forces

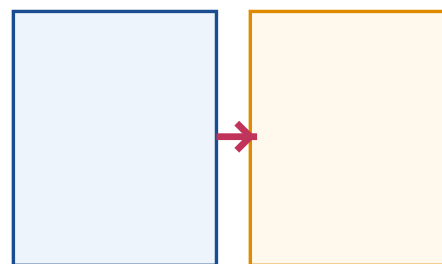
- **Pure fugacity f_a** – escaping tendency set by a–a (like) attraction.
- **Henry's constant H_a** – escaping tendency from a–b (unlike) interaction at infinite dilution.

Case 1 ($f_a < H_a$): a–a attraction $>$ a–b $\Rightarrow$ lower pure-state escaping tendency.

Case 2 ($f_a > H_a$): a–b attraction stronger than a–a.

Liquid α

Vapor β



$f_\alpha = f_\beta$ at equilibrium

Module 5 resource: LearnChemE – Fugacity Criteria & Intermolecular Mechanics

6 EXCESS PROPERTY FORMULATIONS & ACTIVITY MODELS

Excess property: $M^E = M - M^{id}$ (18)

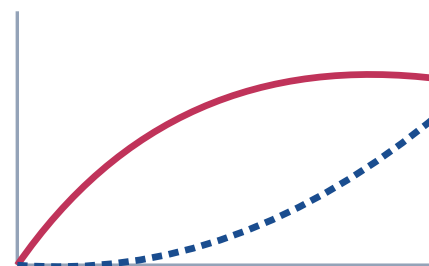
Activity coefficient:

$$f_i^I = x_i \gamma_i f_i^I, \quad RT \ln \gamma_i = G_i^E \quad (19)$$

1-Param Margules: $G^E/RT = A x_1 x_2 \Rightarrow \ln \gamma_1 = A x_2^2, \ln \gamma_2 = A x_1^2$

2-Param Margules: $G^E/RT = x_1 x_2 (A_{21} x_1 + A_{12} x_2)$

van Laar: $\ln \gamma_1 = A'_{12} (1 + A'_{12} x_1 / A'_{21} x_2)^{-2}$ (20–22)



activity coefficient vs composition

Module 6 resource: NPTEL – Activity Coefficient Models

Why it matters: $\gamma_i(x)$ from these models feeds directly into VLE and azeotrope predictions in Module 7 — the non-ideality "correction factor" on Raoult's law.

7 PHASE EQUILIBRIA (VLE, LLE, VLLE)

7.1 Gibbs Phase Rule

$$F = C - \pi + 2 \quad (23)$$

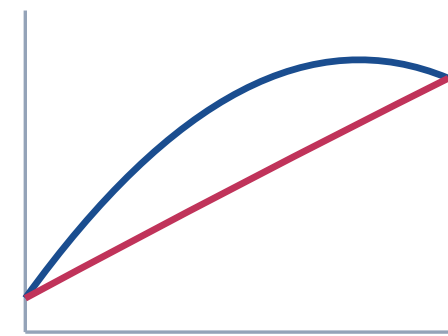
7.2 VLE Calculations

γ - ϕ rigorous equilibrium:

$$y_i \phi_i P = x_i \gamma_i P_i^{\text{sat}} \phi_i^{\text{sat}} \exp[V_i^L(P - P_i^{\text{sat}})/RT]$$

Low-pressure modified Raoult's law:

$$y_i P = x_i \gamma_i P_i^{\text{sat}} \quad (24-25)$$



T-x-y diagram (bubble & dew curves)

7.3 Azeotropic Systems

Minimum-boiling (positive deviation): $\gamma_i > 1$, $G^E > 0$ – unfavorable unlike interactions.

Maximum-boiling (negative deviation): $\gamma_i < 1$, $G^E < 0$ – unusually strong unlike interactions.

Module 7 resource: LearnChemE – VLE & Azeotrope Screencasts

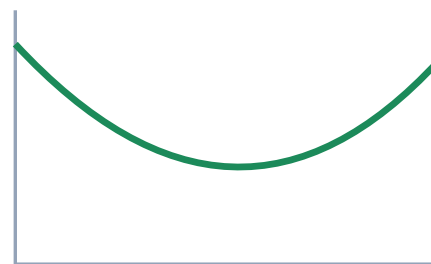
8 CHEMICAL REACTION EQUILIBRIUM

Stoichiometry: $\sum \nu_i A_i = 0$. Mole fraction vs extent ξ :

$$y_i = (n_{i,0} + \nu_i \xi) / (n_{0,\text{tot}} + \nu_{\text{tot}} \xi) \quad (26)$$

Standard Gibbs energy & equilibrium constant:

$$\Delta G^\circ = -RT \ln K_{\text{eq}}(T) \Rightarrow K_{\text{eq}} = \prod_i (\hat{a}_i)^{\nu_i} = (P/P^\circ)^{\nu_{\text{tot}}} \prod_i y_i^{\nu_i} \quad (27)$$



ln K
eq vs 1/T (van't Hoff)

8.2 van't Hoff Equation

$$d(\ln K_{\text{eq}})/dT = \Delta H^\circ(T)/RT^2 \quad (28)$$

Module 8 resource: NPTEL – Reaction Equilibrium Formulations

9 CASE STUDY – WATER-GAS SHIFT REACTION (WGSR)

9.1 Reaction Fundamentals & Pressure Independence

Reaction: $\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$

Stoichiometric sum: $v_{\text{tot}} = (-1) + (-1) + 1 + 1 = 0$.

Since $v_{\text{tot}} = 0$, $(P/P^\circ)^{v_{\text{tot}}} = 1$ — equilibrium extent is

independent of total pressure P .



9.2 Quantitative Reaction Equimodels

Condition A – Equimolar feed (1 mol CO, 1 mol H₂O, 1100 K, 1 bar):

$n_{\text{CO}} = 1 - \xi$, $n_{\text{H}_2\text{O}} = 1 - \xi$, $n_{\text{CO}_2} = \xi$, $n_{\text{H}_2} = \xi$

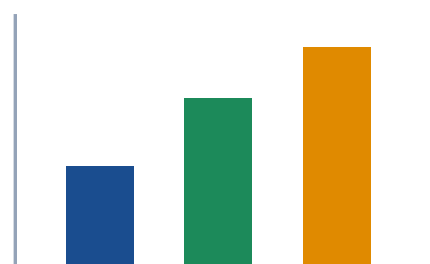
$$K_{\text{eq}} = \xi^2 / (1 - \xi)^2 \quad (29)$$

Condition B – Excess steam feed (1 mol CO, 2 mol H₂O):

$n_{\text{CO}} = 1 - \xi$, $n_{\text{H}_2\text{O}} = 2 - \xi$, $n_{\text{CO}_2} = \xi$, $n_{\text{H}_2} = \xi$

$$K_{\text{eq}} = \xi^2 / [(1 - \xi)(2 - \xi)] \quad (30)$$

Excess steam drives ξ forward → higher CO conversion.



CO conversion: equimolar < excess steam

Module 9 resource: LearnChemE – Water-Gas Shift Reaction Solutions

Previous Year Questions (PYQs)

Useful for practice and revision — compiled question bank for CHE221:

<https://drive.google.com/drive/folders/1PaKx8gxKWZPyCMtWRW2YSTnmme6kmhs0>

Exam-prep checklist

- Derive all four Maxwell relations from memory (the mnemonic square)
- Be able to switch between γ - ϕ and modified Raoult's law VLE forms
- Practice K_{eq} extent-of-reaction problems for $v_{\text{tot}} = 0$ and $\neq 0$ cases
- Know when Bi, Fo-style simplifications (ideal solution, ideal gas) are valid

Formula quick-index

- Eq. (9–12): Maxwell relations
- Eq. (17): Fugacity equality
- Eq. (24–25): VLE (γ - ϕ , Raoult's)
- Eq. (27–28): K_{eq} , van't Hoff

Core references: Smith, Van Ness & Abbott, 8e · Callen, 2e · Koretsky, 2e · Prausnitz/Lichtenthaler/de Azevedo, 3e · NPTEL (IIT Madras) · MIT OCW 5.60 · LearnChemE Screencasts